Importing Data in R: Exploring Various Packages and Their Advantages: Data Analysis for Decision-Making (SS 2026)

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Abstract

This document describes the importance of importing data in R, introducing the packages and their advantages.

1 Why Do We Import Data?

2 Why Do We Import Data? Why with R?

- **Accessing the "Raw" Reality**
- **The Reproducibility Imperative**
- **Auditable**
- **Efficiency and Automation**
- **Handling Complexity**

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3 Different Data Types

1. **Rectangular/Tabular Data**
2. **Proprietary Spreadsheets**
3. **Statistical Software Files**
4. **Relational Databases**
5. **Spatial Data**
6. **Base Text and Fixed-Width Files**
7. **Scientific/Specialized Data**

4 Graphical and Programmatic Implementation by Data Type

4.1 Standard Tabular Data (.csv, .tsv)

- **Graphical:** Use the RStudio "Import Dataset" wizard and select "From Text (readr)". This allows for a live preview where you can visually change delimiters, skip rows, and rename columns while the GUI generates the code automatically Luraschi (2023).

- **Code Wickham and Grolemund (2016)** :

```
library(readr)
df_readr <- read_csv("data.csv")
df_readr
```

large CSV:

```
library(data.table)
df_fread <- fread("bigdata.csv")
df_fread
```

4.2 Proprietary Excel Files (.xls, .xlsx)

- **Graphical:** Select "From Excel**"** in the RStudio wizard. This is crucial for navigating multi-sheet workbooks, as you can select specific sheets and define a cell range (e.g.,"A1:E20") graphically Luraschi (2023).

- **Code** Chapman University (2026):

```
library(readxl)
df_excel <- read_excel("file.xlsx")
df_excel
```

4.3 Statistical Software Files (.sav, .sas7bdat, .dta)

- **Graphical:** Select "From SPSS/SAS/Stata" in RStudio. This wizard is particularly helpful as it shows how proprietary labels and missing value codes will be translated into R factors Luraschi (2023).

- **Code R Core Team (2015):**

```r
library(haven)
# Import SPSS file
df_spss <- read_sav("survey_data.sav")
# Import Stata file
df_stata <- read_dta("economic_data.dta")
# Preview imported data
head(df_spss)
head(df_stata)
```

4.4 Relational Databases (.db, .sqlite, .mdb)

- **Graphical:** Access through the RStudio "Connections" pane. Once a DBI connection is established, you can browse table schemas and preview data before importing Luraschi (2023).
- **Code R Core Team (2015):**

```r
library(DBI)
library(RSQLite)
# Connect to SQLite database
con <- dbConnect(RSQLite::SQLite(), "company.db")
# Read table
df_db <- dbGetQuery(con, "SELECT * FROM employees")
# Show data
df_db
# Disconnect
dbDisconnect(con)
```

4.5 Spatial / GIS Data (.shp, .tif, .kml)

- **Graphical:** Usually handled programmatically, though GIS software like GRASS can be linked to R to browse spatial layers.
- **Code Bivand et al. (2008) :**

```r
library(sf)
# Read shapefile
spatial_data <- st_read("sample_shapefile.shp")
# Preview spatial object
print(spatial_data)
```

4.6 Base Text and Fixed-Width Files (.txt, .dat, .fwf)

- **Graphical:** Use the "From Text (base)" option in the RStudio wizard for legacy text formats that require manual field-width definitions Luraschi (2023).
- **Code R Core Team (2015):**

```r
# Import fixed-width file
df_fwf <- read.fwf(
  "legacy_data.fwf",
  widths = c(3, 6, 5),
  col.names = c("ID", "Name", "Score")
```

```
)

df_fwf
```

4.7 Scientific Spectral Data (.jdx, .abs, .cnv, .nc)

- **Graphical:** Utilize specialized package GUIs or high-level functions that automate folder-wide imports Gruson et al. (2019).

- **Code Gruson et al. (2019) :**

```
library(lightr)

# Import spectral data
spec_data <- lr_get_spec("spectrum.jdx")

# Show spectral data
head(spec_data)
```

4.8 Cloud Sheets (Google Sheets/ URLs)

- **Code: Chapman University (2026)**

```
library(googlesheets4)
# Import Google Sheet
gs4_deauth()
sheet_data <- read_sheet(
  "https://docs.google.com/spreadsheets/d/19Z6SowrQtCS_NsIAZ9ZcaZeb532cSlOOTPCSqfKKsSw/e
)

sheet_data
```

4.9 Hierarchical Data (.json)

- **Graphical:** While RStudio does not have a dedicated "JSON Wizard" in the same way it does for Excel or CSV, users can often preview JSON structures through the File Pane or by using specialized viewing packages.

- **Code R Core Team (2015) :**

```
library(jsonlite)
df_json <- fromJSON("data.json")
df_json
```

5 The Comparison Between Packages

- Tabular Data: Very fast import; creates tibbles / Limited by RAM.

- Excel Sheets: No Java required; supports multiple sheets / Limited scalability.

- Statistical Files: Preserves labels and metadata / Limited newest-version support.

- Databases (SQL): Handles large datasets / Requires SQL knowledge.

- Spatial / GIS: Supports CRS metadata / Complex for beginners.
- Hierarchical (Web): Standard for APIs and JSON / Nested data can be complex.
- Batch Import: Automates multi-file import / Requires functional programming.
- Cloud Services: Direct cloud access / Requires OAuth authentication.

6 Class Assignment

Download the files from here link.

6.1 Help The HR Manager!

1. You are helping a HR manager in the company. She has given you CSV file about the personnel information (off-course after signing relative Non-disclusure agreement!). This data file is changing every month and it is a real pain for her to create the report every time! She knows you have R knowledge! and ask you to prepare a dynamic file by import the data into R, so she can easily prepare report very quickly in the beginning of every month. go ahead and do first step which is importing the exe1.csv file into R Studio.

6.2 Everyday Sales Report!

2. You are a manager in a company and the sale department gives you sales reports in excel file every morning. consider that you have recieved exe2.xlsx file today. Import it to R Studio.

6.3 Spare Parts Sheet

3. Imagine that you and your colleagues in the company are working on a google spreadsheet, together. now you want to import data from it into R. Use this Link and import it in the R Studio. if you had problem with the link to Google spreadsheet, upload it (it is available in the exercise zip folder) in your own account, and then use your own link.

7 References

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